

Tanvi Batchu

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EDUCATION

University of Waterloo: Double Major in Computer Science and Finance

Bachelors of Computing and Financial Management

Waterloo, ON

Sept 2025 - Present

EXPERIENCE

Algo Research + Engineering

May 2026 - Dec 2026

RBC Capital Markets

Toronto, ON

- Created **multi-asset algo trading tools**, including optimizing cost for **futures contract routing, ETF to single security mapping** for cost analysis, extending coverage to EMEA, AMRS, and APAC regions
- Developed an **arrival cost decomposition model** isolating order execution performance from external market volatility, publishing research findings to the trade floor and building supporting software tools for trade-wide usage
- Developed and **published company-wide research** on short-term volume prediction, reducing **execution costs by 3%**
- Increased adoption of Algo Monocle from **7 to 30 users** on the RBC Capital Markets trade floor: an **LLM-powered algo trading & research** tool that enabled traders and quants to retrieve **execution data, construct charts, and generate client** reports within a single interface, later extending coverage to EMEA markets
- Engineered **kdb+/q scripts** to parse large timeseries datasets, extracting order data matched to their sector ETF data from trade databases, reducing order data **preparation time from 2 hours to < 10 minutes and report review by 40%**

Quantitative Researcher

Jan 2026 - Present

Marble Investments

Canada

- Analyzed short-term momentum** in “hype stocks” to assess whether abnormal volume spikes and rapid price acceleration produced consistent, **repeatable trading patterns across market conditions**
- Backtested systematic trading signals** incorporating liquidity and volatility thresholds to establish entry & exit criteria
- Evaluated performance using Sharpe ratio, drawdown depth, win-rate consistency, and regime analysis; iteratively **refined signal parameters** to enhance **risk-adjusted returns** and **reduce overfitting**

Project Lead + ML Engineer

May 2026 – Present

WatStreet – Strategy Optimization & World Order Book

Waterloo, ON

- Led a team building a **backtesting framework** comparing **VWAP/TWAP execution algorithms** on major stocks, training a model to recommend **optimal algorithm selection** by market condition
- Built a **world model-based RL system** for LOB execution: learning a latent representation of order book dynamics to simulate realistic market states, training **DDQN** agents with custom epsilon decay inside the simulation

PROJECTS

MarketMeet Robo-Advisor | *Python, YFinance, SciPy, Pandas, NumPy, Matplotlib, Git, Jupyter Notebook*

- Designed a **multi-factor screening model** using **returns, volatility, beta, covariance, and correlation** against TSX/S&P 500 benchmarks to evaluate tracking performance; **Matched market to 0.05% accuracy**
- Applied a **modified Markowitz mean-variance framework** to generate portfolios, adding constraints on volatility
- Created a **backtesting system to simulate performance** across historical periods and **measure Sharpe ratio, drawdowns, tracking error, and cumulative returns**
- Designed a rebalancing process to systematically **adjust allocations based on updated risk and return signals**

LandLock: AI-Powered Risk Intelligence Platform | *Python, FastAPI, React, TypeScript, CrewAI, LangGraph, Mapbox*

- Architected **multi-agent system with CrewAI & LangGraph** to orchestrate 3 **specialized agents** (data validation, risk analysis, mitigation strategy) analyzing **20+ BC government datasets** for climate risk assessment
- Built **FastAPI backend with 15+ RESTful endpoints** managing automated scraping pipelines and file fact extraction
- Integrated **Beautiful Soup** for web scraping municipal data and **OpenAI GPT-4 for LLM-enhanced insights**
- Implemented **test suite (40+ tests)** using **Jest & Pytest** for reliability across agent workflows & data processing

AWARDS

European Girls' Olympiad in Informatics (EGOI) National Camp Qualifier | *CEMC*

Top 10 female competitive programmers out of 4k+ based on Canadian Computing Contest (CCC) scores

2025

Provincial Champion and Top 8% Internationally (ICDC) x2 | *DECA*

2024 & 2025

TECHNICAL SKILLS

Quantitative & Programming: Python (Pandas, NumPy), kdb+/q, C/C++, SQL, R, JavaScript, HTML/CSS

Financial Modeling & Machine Learning: Regression, Classification, Time-Series Analysis, Neural Networks, SHAP, MLFlow

Market Data & Analytics: Financial Data Processing, REST APIs, YFinance, Statistical Analysis, Data Visualization

Infrastructure & Tools: Excel, MS Office Suite, Power BI, Tableau, AWS, Azure, Git, Jupyter Notebook